Athul Krishna Girish

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PROFILE

Dual-degree student pursuing B.Tech in Electronics Engineering (VLSI Design & Technology) at Govt. Model Engineering College (MEC), Kerala, and B.Sc (Hons) in Data Science & Artificial Intelligence at IIT Guwahati. Skilled in machine learning, data science, Python programming, embedded systems, and PCB design, with hands-on experience across real-world projects, internships, and national-level competitions including Smart India Hackathon 2025 and Hult Prize Nationals 2026.

EDUCATION

Govt. Model Engineering College, Thrikkakara — Aug 2024 – Present
B.Tech in Electronics Engineering (VLSI Design & Technology) | CGPA: 9.37 — *Kochi, Kerala*

Indian Institute of Technology, Guwahati — Sep 2024 – Present
B.Sc (Hons) in Data Science & Artificial Intelligence | CGPA: 7.47 — *Online*

TECHNICAL SKILLS

Languages: C, Python, Java, HTML, CSS
ML / Data Science: Physics-Informed Neural Networks (PINNs), LSTM, ARIMA, Prophet, Time Series Forecasting, Data Analysis, Data Visualization, Statistical Modelling
Hardware & VLSI: Verilog, VHDL, FPGA Implementation (Xilinx Vivado), LTspice, KiCad, PCB Design, Embedded Systems, IoT
Developer Tools: Git, GitHub, VS Code, Linux, Canva

CERTIFICATIONS

Deep Learning — NPTEL Certification — IIT Ropar

Python for Data Science — NPTEL Certification — IIT Madras

Integrated Circuits, MOSFETs, Op-Amps and their Applications — NPTEL Certification — IISc Bangalore

Digital Design with Verilog — NPTEL Certification — IIT Guwahati

PROFESSIONAL EXPERIENCE

Department of Electronics Engineering, Model Engineering College — FPGA Design Intern — 15-25 June 2026
On-site, Kerala, India

- Worked on FPGA-based digital system design using Verilog, with practical exposure to RTL design, combinational/sequential circuits, Finite State Machines (FSM), and FPGA prototyping.
- Developed an FPGA-Based Intelligent Dual Elevator Scheduling System with Fire Evacuation in a 4-member team, implementing optimized scheduling, smart request allocation, and emergency evacuation logic; demonstrated on FPGA hardware and emerged as the winning team in the final project evaluation.

Zidio Development — Data Science & Analytics Intern — Jun 2025 – Jul 2025
Remote, India

- Developed a full-pipeline stock price forecasting system using Apple Inc. (AAPL) stock data, implementing ARIMA, Facebook Prophet, and LSTM models to capture linear trends, seasonality, and complex temporal patterns in financial data.
- Achieved improved prediction accuracy by comparing model performance metrics; built interactive data visualizations analyzing trends, moving averages, and actual vs. predicted price comparisons.

India Space Lab — Space Tech Intern — Jun 2025 – Jul 2025
Remote, India

- Completed an intensive interdisciplinary training program on next-generation aerospace technologies including UAVs, CubeSats, CanSats, Rocketry, Remote Sensing, GIS, IoT, and Embedded Systems.
- Gained hands-on experience with industry-grade tools: SolidWorks & ANSYS for CAD/CAE, ArduPilot and ROS/Gazebo for UAV flight simulation, and EasyEDA for PCB design.

PROJECTS

SIDDHANT — Astronomical Yantra Dimensioning Platform — SIH 2025 Grand Finale
Team Meta Learners | Problem Statement by Indian Knowledge Systems (IKS), AICTE — *Smart India Hackathon 2025*

- Built a platform generating precise, location-specific dimensions for 14 ancient Indian astronomical Yantras for any coordinate in India, addressing a national-level challenge set by Indian Knowledge Systems (IKS), AICTE.
- Deployed Physics-Informed Neural Networks (PINNs) with sigmoid parameterization and physics projection layers, layered AR visualization of 3D Yantra models, automated government-scraped soil and cost data, and sovereign mapping via MapMyIndia.
- Tech Stack: Python, PINNs, MapMyIndia API, 3D Modeling, AR Visualization, CAD

Project Link: siddhant-project.vercel.app

FPGA-Based Intelligent Dual Elevator Scheduling System with Fire Evacuation 2026

Internship project | Department of Electronics Engineering, Model Engineering College, Kochi, Kerala

- Designed and implemented an FSM-based dual elevator controller in Verilog HDL, enabling intelligent elevator dispatch, optimized floor request allocation, and reduced passenger waiting time through efficient scheduling logic.
- Integrated a fire evacuation safety mode with priority override control for emergency handling; validated the complete design through simulation and live deployment on FPGA Boolean Board, where the project was recognized as the best-performing system in final evaluation.
- Tech Stack: Verilog HDL, FPGA, FSM Design, RTL Design, Digital System Design, Boolean Board

CrowdSense — Real-Time Crowd Intelligence & Safety Platform 2025

- Built a full-stack AI crowd monitoring platform using Google Gemini 2.5 Flash Vision AI to estimate crowd density from live camera feeds, assess risk levels, and deliver real-time safety guidance with zero data storage for complete privacy.
- Engineered a custom capacity-aware risk engine overriding generic AI assessments with venue-specific, and implemented live frame analysis every 3-5 seconds, browser-native push notifications, heatmap overlays, and Aami - a Gemini-powered AI safety assistant.
- Tech Stack: React 18, TypeScript, Tailwind CSS, Framer Motion, Deno Edge Functions, Google Gemini 2.5 Flash, TanStack React Query, Web Push API

Live Demo: crowdsense-live.vercel.app

EXTRACURRICULAR ACTIVITIES

IEEE Circuits and Systems Society (CASS), IEEE Student Branch MEC — Secretary June 2026 – Present

- Spearhead the planning and execution of technical workshops, seminars, and chapter initiatives focused on circuits, systems, and emerging semiconductor technologies.
- Drive team coordination, strategic communication, event management, and documentation, enhancing chapter operations and fostering technical engagement within the student community.

Hult Prize India Nationals 11-12 April 2026

Indian Institute of Technology Bombay

- Secured 1st place in campus qualifiers and represented the college at national finals, pitching Harit Setu, a climate-tech platform designed to bridge smallholder Indian farmers with the carbon credit market through transparent carbon accounting and green value creation.